Business Analysis and Valuation

Cash Flow and Valuation

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FCF Valuation Model

$$Value = \frac{FCF_1}{(1+wacc)} + \frac{FCF_2}{(1+wacc)^2} + \ldots = \sum_{t=1}^{\infty} \frac{FCF_t}{(1+wacc)^t}$$

Value of Company = Present value of cash flows during the explicit forecasting period + Present Value of Cash flows beyond the explicit forecasting period (**terminal value**)

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A few starting problems…

- To calendarize or not?
- Stand-alone financials or consolidated financials?
  - What is the difference?
  - Control is presumed to exist when the parent owns, directly or indirectly through subsidiaries, more than half of the voting power of an entity unless, in exceptional circumstances, …
- Should we use information from equity research reports?
  - By all means, use them… give references …and never go by the recommendations made in such reports as they are mostly biased.
  - They contain useful information about the industry.
  - Should you use Chat-GPT or Claude, Gemini, or Command R+?
    - Use them, if you want. But write the prompts used, the response obtained, and how the same has been used in the report.

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Quarter-end Balance Sheet: Indian Regulations

Regulation / Standard	Applicability	Requirement	Balance Sheet Disclosure
SEBI LODR Regulations (2015)	Listed companies on Indian stock exchanges	Quarterly financial results must be published within 45 days of quarter-end	Condensed balance sheet, profit & loss, cash flow, and notes
Ind AS 34 (Interim Financial Reporting)	Companies mandated to follow Ind AS (mainly listed and large entities)	Interim financial statements must include minimum components	Condensed balance sheet (quarter-end + comparative year-end), P&L, cash flow, notes
AS 25 (Interim Financial Reporting)	Smaller companies not covered under Ind AS	Quarterly/half-yearly reporting	Condensed balance sheet, P&L, cash flow, notes
MCA Quarterly Disclosure (effective April 1, 2026)	Public limited companies (listed + certain unlisted meeting thresholds)	Unaudited quarterly financial statements to be filed on MCA21 portal within 45 days	Balance sheet data mandatory along with revenue, profit, EPS, deviations

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Quarter-end Balance Sheet: Havells

3. Balance Sheet (Abridged)

(In Rs crores)	June'26 (Unaudited)	Mar'26 (Audited)
ASSETS		
Non-current assets		
Property, plant and equipment and intangibles	5,875	5,761
Investments	956	956
Other non-current assets	361	247
Total non current assets	**7,192**	**6,964**
Current assets		
Inventories	5,016	4,398
Financial assets		
(i) Trade receivables	723	782
(ii) Cash and cash equivalents	291	778
(iii) Bank balances other than (ii) above	1,206	1,573
(iv) Other financial assets	4	9
Other current assets	384	221
Total current assets	**7,623**	**7,760**
Total assets	**14,815**	**14,724**

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Standalone financials or consolidated financials?

	P Limited	S Limited	M Limited
P Limited's equity Ownership	-	80%	20%
Net Assets	₹500 million	₹100 million	₹100 million
Net Income	₹100 million	₹30 million	₹40 million
Market Capitalization	₹1,000 million	₹300 million	₹400 million

- Valuation if you start from standalone financials…
- Valuation if you start from consolidated financials…
- Differences
- Which method is better?

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Controlling More than 50% shares

- **Control and Consolidation**: Holding more than 50% of the shares typically signifies that the parent company has control over the subsidiary. When control exists, the parent company is required to prepare consolidated financial statements, presenting the financial position, results of operations, and cash flows of the parent and subsidiary as if they were a single economic entity.
- **Consolidated Financial Statements**: Consolidated financial statements combine the accounts of the parent company and all its subsidiaries into a single set of financial statements. This means that the assets, liabilities, equity, revenues, and expenses of the subsidiary are combined with those of the parent company.
- **Elimination of Intercompany Transactions**: In the consolidation process, intercompany transactions, balances, and profits or losses on transactions between the parent and subsidiary are eliminated.
- **Proportional Ownership**: The portion of the subsidiary's equity that is not owned by the parent company is known as non-controlling interest (NCI) or minority interest. This portion is presented separately in the consolidated financial statements to indicate that there are other owners of the subsidiary.

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When you hold between 20-50% of the shares: Equity Accounting

- **Initial Recording of Investment**: When a company invests in another entity and it meets the criteria for using the equity method (generally a long-term investment representing a stake of 20-50% ownership), the investment is initially recorded at cost, which includes the purchase price plus any directly attributable costs.
- **Proportionate Share of Income/Loss**: After the initial recording, the investing company adjusts the carrying value of the investment to recognize its share of the investee company's earnings or losses. For example, if Company A owns 30% of Company B, then Company A would report 30% of Company B's profits or losses in its own income statement.
- **Adjustments for Dividends**: When the investee company pays dividends to the investing company, the dividends are recorded as a reduction in the carrying value of the investment.

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Subsidiaries of Titan

₹ in crore

Particulars	As at 31st March 2026	As at 31st March 2025
1) Investment in equity instruments (unquoted)		
(i) In subsidiary companies (at cost unless stated otherwise)		
4,70,50,000 (Previous year: 4,70,50,000) fully paid equity shares of ₹ 10 each in Titan Engineering & Automation Limited	235	235
3,33,46,141 (Previous year: 3,33,46,141) fully paid equity shares of ₹ 2 each in CaratLane Trading Private Limited	5,202	5,202
36,701 (Previous year: 36,701) fully paid equity shares of AED 1,000 each in Titan Holdings International FZCO	84	84
2,96,500 (Previous year: 2,96,500) fully paid equity shares of USD 100 each in TCL North America Inc	232	232
1,50,00,000 (Previous year: 1,50,00,000) fully paid equity shares of ₹ 10 each in Titan Commodity Trading Limited	15	15
	5,768	**5,768**
(ii) In associate company (at cost unless stated otherwise)		
15,00,000 (Previous year: 15,00,000) fully paid equity shares of ₹ 10 each in Green Infra Wind Power Theni Limited	2	2
	5,770	**5,770**

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Market Value of Investments

Particulars	As at 31st March 2026	As at 31st March 2025
iii) Investments in non-convertible debentures carried at amortised cost - unquoted		
Investment in non convertible debentures	972	469
Investment in Government Securities	66	118
	1,038	**587**
Aggregate value of investments	**6,836**	**6,386**
Aggregate book value of quoted investments	**2**	**3**
Aggregate market value of quoted investments	**2**	**3**
Aggregate book value of unquoted investments	**6,834**	**6,383**
Aggregate amount of impairment in value of investments	**-**	**-**

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Example of Grasim

₹ in crore

Particulars	Place of Incorporation	Face Value	Number of Shares	Grasim's Ownership %	As at 31st March 2025	Number of Shares	Grasim's Ownership %	As at 31st March 2024
Equity Shares								
Quoted:								
Subsidiaries: Carried at Cost								
UltraTech Cement Limited	India	₹ 10	16,53,35,150	56.11%	2,636.25	16,53,35,150	57.27%	2,636.25
Aditya Birla Capital Limited	India	₹ 10	1,36,98,09,351	52.54%	18,846.95	1,36,98,09,351	52.68%	18,846.95
					21,483.20			**21,483.20**
Unquoted:								
Subsidiaries: Carried at Cost								
ABNL Investment Limited	India	₹ 10	2,81,40,000	100.00%	108.79	2,81,40,000	100.00%	108.79
Samruddhi Swastik Trading and Investments Limited	India	₹ 10	81,66,666	100.00%	26.50	65,00,000	100.00%	6.50
Aditya Birla Renewables Limited	India	₹ 10	85,57,19,413	100.00%	849.42	83,30,51,406	100.00%	826.75
ABReL Solar Power Limited	India	₹ 10	7,36,80,817	26.00%	73.68	7,10,83,886	26.00%	71.09
Grasim Business Services Private Limited	India	₹ 10	1,00,000	100.00%	0.10	1,00,000	100.00%	0.10
					1,058.49			**1,013.23**
Joint Ventures: Carried at Cost								
AV Group NB Inc., Class 'A' Shares of aggregate value of Canadian Dollar 38.25 million	Canada	WPV	2,04,750	45.00%	153.04	2,04,750	45.00%	153.04
Aditya Birla Elyaf Sanayi Ve Ticaret Anonim Sirketi	Turkey	TL 10	16,665	33.33%	0.47	16,665	33.33%	0.47
AV Terrace Bay Inc.	Canada	CAD 1	8,16,00,000	40.00%	484.49	3,80,00,000	40.00%	218.24
Less: Provision for impairment in the value of Investment {Note 3.10 (iii)}					(484.49)			(218.24)
Aditya Group AB	Sweden	SEK 1000	50	33.33%	274.89	50	33.33%	274.89
Bhubaneswari Coal Mining Limited	India	₹ 10	3,35,40,000	26.00%	33.54	3,35,40,000	26.00%	33.54
Aditya Birla Power Composites Limited	India	₹ 10	2,27,32,740	51.00%	22.73	2,27,32,740	51.00%	22.73
Birla Jingwei Fibres Company Limited, Shares of aggregate value of RMB 174.53 million	China	WPV	-	26.63%	117.40	-	26.63%	117.40
Birla Advanced Knits Private Limited	India	₹ 10	2,50,00,000	50.00%	25.00	2,50,00,000	50.00%	25.00
Less: Provision for impairment in the value of Investment {Note 3.10 (i)}					(25.00)			-
					602.07			**627.07**

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Maintaining Consistency in Assets & Liabilities & Cash Flows & WACC

- Start with Liabilities (find WACC).
  - Remember to make a note of liabilities that were excluded while computing WACC.
- Divide assets into operating and non-operating assets. It has nothing to do with how accountants divide assets into operating and non-operating assets. It depends on your valuation objective.
  - FCF must be generated from operating assets only.
  - Remember to add back market value of non-operating assets at the end to PV of FCF to obtain the enterprise value.

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Understanding the different line items…

- Line items may be different in service-oriented companies.
- **Revenue**
  - **Sales or Net Sales?**
  - Oyo Rooms used to charge 22% commission from the hotel partners
    - **Interglobe Aviation**

Standalone Statement of Profit and Loss

for the year ended March 31, 2023

(Rupees in millions, except for share data and if otherwise stated)

Particulars	Note	For the year ended March 31, 2023	For the year ended March 31, 2022
Income			
Revenue from operations	22	544,464.53	259,309.27

28. Other expenses

Particulars	For the year ended March 31, 2023	For the year ended March 31, 2022
Repairs and maintenance	1,851.44	1,381.90
Insurance		
- aircraft	1,005.49	752.52
- others	665.40	764.03
Ground handling charges	11,443.89	7,507.47
Reservation cost	2,159.87	1,383.09
Commission	6,843.33	2,672.66
Sales promotion and advertisement	879.05	586.63

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May need to restate previous years' figures

- We need a base number to forecast future sales.
  - Walmart International revenue fell from $121.4bn in 2021 FY to $101bn in 2022 FY.
  - It sold Asda (UK) and Seiyu (Japan) during the year.

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Spreading Revenue Data

India Cement & Materials — sector deep-dive

India cement crossed ~490 MT in FY26 (+9% YoY) — its largest demand year on record — while the sector's most...

VestAI Research Updated 2026-07-06 14 min read 24 companies

TL;DR — India cement crossed ~490 MT in FY26 (+9% YoY) — its largest demand year on record — while the sector's most... Key figures: ~490 MT Cement demand FY26 (+9% YoY, record) and ~660 MTPA Industry capacity; ~72% utilisation FY26. This report screens every listed Cement & Materials name on cash conversion, balance-sheet quality and valuation.

~490 MT	~660 MTPA	₹87,384 cr	73.7 MT	160-170 MT
Cement demand FY26 (+9% YoY, record)	Industry capacity; ~72% utilisation FY26	UltraTech revenue FY26 (+17%); EBITDA ₹17,598 cr +32%	Ambuja record volume FY26; EBITDA +31% to ₹6,539 cr	New cement capacity FY26-28 — biggest wave since 2013

Source: OrionAI from vestai.io (https://www.vestai.io/chat)

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Spreading Revenue Data: India Cements

Metric	Reported Data	Teaching Notes
Installed Capacity	~15 MTPA (India Cements)	Industry total ~660 MTPA.
Capacity Utilization	70% (FY26)	Improved from 62% in FY25.
Cement Sold	104.5 lakh tonnes (10.45 MT)	Up from 8.98 MT in FY25.
Average Realization	~₹4,285 per tonne (back-calculated)	Revenue ÷ volume.
Revenue	₹4,485 crore	Up from ₹4,080 crore in FY25.
EBITDA per Tonne	~₹497 (Q4 FY26)	Shows margin improvement.
PAT	₹65 crore	Turnaround from ₹656 crore loss in FY25.
Capex Plan	₹2,014 crore over 2 years	Expansion by 2.8 MTPA.

Source: vestai.io

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Spreading Revenue Data: Walmart

Metric	Reported Data (FY2026)	Teaching Notes
Total Stores	~10,900 (U.S. ~4,600, International ~5,700, Sam's Club ~600) Yahoo Finance	Base for per-store revenue modeling.
Total Retail Square Footage	~1,057 million sq ft Yahoo Finance	Includes U.S., International, and Sam's Club.
Total Revenue	$713.2 billion Yahoo Finance	Consolidated net sales.
Revenue per Store	$713.2B ÷ 10,900 ≈ **$65M per store annually**	Useful for store-level forecasting.
Revenue per Square Foot	$713.2B ÷ 1,057M ≈ **$675 per sq ft annually**	Key retail efficiency metric.
Segment Breakdown	U.S.: $483B International: $130.4B Sam's Club: $93B Yahoo Finance	Shows geographic contribution.
Comparable Store Sales Growth	U.S. +4.1% YoY International +18% YoY Yahoo Finance	Indicates organic growth beyond new square footage.

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Spreading Revenue Data: Meta

Total Accounts	~4.1 billion monthly active people (Family of Apps)	Includes Facebook, Instagram, WhatsApp, Messenger, Threads.
Daily Active Users (DAP)	3.56 billion (March 2026)	+4% YoY growth.
Average Time Spent	~60 minutes/day per user (industry estimates)	Engagement varies by geography and age group.
Total Daily Minutes	3.56B × 60 = ~213B minutes/day	This is the "attention pool" Meta monetizes.
Quarterly Revenue	$56.3 billion (Q1 2026)	~33% YoY growth.
Revenue per Minute	$56.3B ÷ (213B × 90 days) ≈ **$0.0029 per minute**	Shows how each minute of attention converts to ad dollars.
Revenue per User per Day	$56.3B ÷ (3.56B × 90) ≈ **$0.52 per user/day**	Equivalent to ~$15.6 per user/month.
Revenue Mix	~98% ads, ~2% Reality Labs	Reality Labs revenue ~$3–4B annually, but loss-making.

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Spreading Revenue Data: Anthropic

Step	Assumptions / Data	Result
Corporate Clients	~300,000 business customers (including ~1,000 large enterprises)	Base for segmentation
Token Usage per Client (monthly)	Enterprise: ~40M Mid-market: ~5M SMB: ~0.5M	Segmented usage
Total Tokens Used (monthly)	~400 billion	Back-solved from reported revenue
Average Price per Million Tokens	~$10 blended (across Opus, Sonnet, Haiku)	Weighted average
Monthly Revenue	400B ÷ 1M × $10 ≈ $3.9B	Matches reported trajectory
Annualized Revenue	~$47B	Consistent with disclosed run rate

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Spreading Revenue Data: Titan

Metric	Reported Data	Teaching Notes
Total Stores	~3,603 (India + International)	Includes Tanishq, Mia, Zoya, CaratLane, Titan World, Fastrack, Helios, Titan Eye+, etc.
Jewellery Stores	~1,194 (Tanishq, Mia, Zoya, CaratLane)	Jewellery is ~80% of Titan's revenue.
Watches & Wearables Stores	~1,310	Titan, Fastrack, Helios, premium formats.
Eyecare Stores	~840 (Titan Eye+)	Growing double-digit annually.
Emerging Businesses	~96	Fragrances, bags, sarees.
Total Retail Space	~10.5 million sq ft (estimate)	Jewellery stores are larger formats; watches/eyecare smaller.
Total Revenue	₹76,078 crore	Consolidated, ex-bullion.
Revenue per Store	₹76,078 Cr ÷ 3,603 ≈ **₹21.1 Cr per store annually**	Jewellery stores skew higher (~₹50–60 Cr/store).
Revenue per Square Foot	₹76,078 Cr ÷ 10.5M sq ft ≈ **₹7,240 per sq ft annually**	Jewellery formats drive higher productivity.
Segment Revenue	Jewellery: ₹59,463 Cr CaratLane: ₹4,702 Cr Watches: ₹5,105 Cr Eyecare: ₹1,452 Cr Emerging: ₹508 Cr International: ₹2,734 Cr	Jewellery dominates Titan's portfolio.

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Other Operating Income

- Questions to be asked…
  - Is this part of the main business activity of the company?
  - Is this a recurring income?

		Is this part of the main business activity of the company?	
		Yes	No
Is this a recurring income?	Yes	Treat as Operating Income	Treat as Non -operating Income
	No	Ignore	Ignore

Examples (Operating or non-operating?:
- Income from credit card operations by a retail firm
- Drug retailer gets revenue from selling containers
- Income from Investment Income (trade vs. non-trade)
- Gain on sale of assets

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Bajaj Auto: Other Income

Statement of Profit and Loss

(₹ In Crore)

Particulars	Note No.	For the year ended 31 March 2025	2024
Revenue from contracts with customers		48.247.22	43,578.87
Oth			1,106.36
Rev			44,685.23
Oth			1,402.45
Tot			**46,087.68**

24 Other income

(₹ In Crore)

Particulars	For the year ended 31 March 2025	2024
Investment income		
Interest income on fixed income securities	123.81	195.55
Amortisation of (premium)/discount on acquisition of fixed income securities	1.35	96.50
Interest income on fixed deposits	87.29	141.85
Interest income on loans	7.64	–
Interest income on exchange traded funds	436.49	443.19
Interest income on fixed maturity plans	49.55	46.90
Interest income from financial assets including amortised cost	706.13	923.99
Dividend income on other strategic investments	30.10	43.05
Gain on valuation and realisation of mutual funds measured at FVTPL	633.28	437.11
Profit/(loss) on sale of other investments, net, measured at amortised cost	31.05	(16.61)
	1,400.56	1,387.54
Others		
Surplus on sale of property, plant and equipment	3.23	5.83
Interest on tax refunds/credits	17.15	–
Gains on exchange fluctuations	–	9.08
	20.38	14.91
	1,420.94	1,402.45

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Assets of Bajaj Auto

Balance Sheet

(₹ In Crore)

Particulars	Note No.	As at 31 March 2025	2024
ASSETS			
Non-current assets			
Property, plant and equipment	2	3,500.17	3,137.85
Capital work-in-progress	2	26.02	24.29
Investment property	3	47.78	48.90
Intangible assets	4	2.81	11.91
Intangible assets under development	4	2.24	3.20
Investments in subsidiaries	5A	4,201.95	2,088.42
Financial assets			
Investments	5B	18,784.83	17,524.60
Loans	6	1,082.14	2.06
Other financial assets	7	29.38	27.68
Income tax assets (net)		1,237.61	905.74
Other non-current assets	8	86.66	118.82
		29,001.59	**23,893.47**
Current assets			
Inventories	9	1,957.90	1,695.62
Financial assets			
Investments	5B	5,583.45	3,837.98
Trade receivables	10	2,282.64	2,122.40
Cash and cash equivalents	11	813.42	448.61
Other bank balances	12	515.13	1,129.51
Loans	6	2.55	3.21
Other financial assets	7	1,486.95	463.90
Other current assets	8	785.27	664.13
		13,427.31	**10,365.36**
		42,428.90	**34,258.83**

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Definition

Total Operating Income = Net Sales + Other Operating Income

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Operating Expenses (Opex) line items…

- Raw materials
- Wages and salaries
- Other manufacturing expenses
- Selling and distribution expenses
- Administrative expenses
- Other expenses (put all smaller items here)
- Depreciation
- Compute the % of total sales (or fixed assets) over the last 3-5 years
- ***Compute EBITDA and EBITDA Margin***

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Opex (excluding Depreciation) as % of Revenue

	2019	2020	2021	2022	2023
Revenue	19,069.97	20,009.64	20,602.00	27,210.00	38,270.00
Raw Materials	13,987.18	14,511.81	15,769.00	20,658.00	29,046.00
% of Revenue	*73.3%*	*72.5%*	*76.5%*	*75.9%*	*75.9%*
Employee Benefit Expenses	878.79	1,040.12	911.00	1,143.00	1,362.00
% of Revenue	*4.6%*	*5.2%*	*4.4%*	*4.2%*	*3.6%*
Other Expenses	2,201.79	2,039.98	2,221.00	2,130.00	3,092.00
% of Revenue	*11.5%*	*10.2%*	*10.8%*	*7.8%*	*8.1%*
EBITDA	2,002.21	2,417.73	1,701.00	3,298.00	4,796.00
EBITDA Margin	*10%*	*12%*	*8%*	*12%*	*13%*

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Forecasting Depreciation

	2019	2020	2021	2022	2023
Revenue	19,069.97	20,009.64	20,602.00	27,210.00	38,270.00
EBITDA	2,002.21	2,417.73	1,701.00	3,298.00	4,796.00
EBITDA Margin	*10%*	*12%*	*8%*	*12%*	*13%*
Depreciation	138.89	309.69	331.00	347.00	364.00
Net Fixed Assets	1,095	2,067	1,984	1,996	2,288
Depreciation Rate		*28%*	*16%*	*17%*	*18%*

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Definition

Net Operating Income (NOI)
= Total Operating Income
- Total Operating Expenses

	2019	2020	2021	2022	2023
Revenue	19,069.97	20,009.64	20,602.00	27,210.00	38,270.00
EBITDA	2,002.21	2,417.73	1,701.00	3,298.00	4,796.00
Net Operating Income	1,863.32	2,108.04	1,370.00	2,951.00	4,432.00
Operating Margin	*9.8%*	*10.5%*	*6.6%*	*10.8%*	*11.6%*

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Estimating NOPAT (Net Operating Profit Adjusted for Tax)

- Also known as NOPLAT by some consulting firms
  - Net Operating Profit Less Adjusted Tax
- For historical financials, use the cash tax rate. For projections, use the current statutory tax rate.

	2019	2020	2021	2022	2023
Revenue	19,069.97	20,009.64	20,602.00	27,210.00	38,270.00
EBITDA	2,002.21	2,417.73	1,701.00	3,298.00	4,796.00
Net Operating Income	1,863.32	2,108.04	1,370.00	2,951.00	4,432.00
PBT	1,927	2,105	1,233	2,932	4,465
Tax Expenses	602	552	351	779	1,140
Tax Rate	31%	26%	28%	27%	26%
NOPAT	1,281.32	1,555.40	980.00	2,166.95	3,300.43

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Definition: Gross Cash Flow

Gross Cash Flow =

NOPAT + Depreciation

Why is depreciation alone added back?

	2019	2020	2021	2022	2023
NOPAT	1,281.32	1,555.40	980.00	2,166.95	3,300.43
Add: Depreciation	138.89	309.69	331.00	347.00	364.00
Gross Cash Flow	1,420.21	1,865.09	1,311.00	2,513.95	3,664.43

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Estimating Gross Investment

- Computing Capex from financial statements
- Computing increase in operating working capital
  - Operating and Non-Operating Cash
- Items to include in current liabilities
  - What about short-term debt?
- Working Capital for Bharti Airtel as at 31 March 2022
  - Inventory: 4 Million
  - Receivables: 51,525
  - CMS: 3,303
  - Payables and Deferred Revenue: 303,018

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Gross Investment of Titan

	2019	2020	2021	2022	2023
Net Fixed Assets	1,095	2,067	1,984	1,996	2,288
Change in Net Fixed Assets		972	-83	12	292
Add: Depreciation		138.89	309.69	331.00	347.00
Capex		1,111.16	226.44	343.00	639.00

Capex will often be lumpy for most companies.
While forecasting capex, try to link it to your revenue drivers.

Forecasting Capex for Titan:
- Link to number of stores planning to set up.
- If the company plans to increase outsourcing, then it will also affect its capex.

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Operating Working Capital of Titan

	2019	2020	2021	2022	2023
Revenue	19,069.97	20,009.64	20,602.00	27,210.00	38,270.00
Current Assets					
Inventories	6,719.18	7,740.62	7,984.00	12,787.00	14,952.00
% of Revenue	*35%*	*39%*	*39%*	*47%*	*39%*
Trade Receivables	358.23	214.44	291.00	495.00	908.00
% of Revenue	*2%*	*1%*	*1%*	*2%*	*2%*
CMS	1,001.00	356.00	512.00	1,049.00	792.00
Other Receivables	937.63	1,149.76	961.00	2,033.00	1,891.00
% of Revenue	*5%*	*6%*	*5%*	*7%*	*5%*
Total Current Assets	9,016.04	9,460.82	9,748.00	16,364.00	18,543.00
Operating Current Assets	8,015.04	9,104.82	9,236.00	15,315.00	17,751.00
% of Revenue	*42%*	*46%*	*45%*	*56%*	*46%*
Current Liabilities					
Borrowings	2,287.63	2,302.62	4,272.00	5,579.00	6,503.00
Trade Payables	772.34	509.94	695.00	1,055.00	965.00
% of Revenue	4%	3%	3%	4%	3%
Other Current Liabilities	2,109.28	2,431.32	2,226.00	2,925.00	4,053.00
% of Revenue	11%	12%	11%	11%	11%
Total	5,169.25	5,243.88	7,193.00	9,559.00	11,521.00
Operating Current Liabilities	2,881.62	2,941.26	2,921.00	3,980.00	5,018.00
% of Revenue	15%	15%	14%	15%	13%
Operating Working Capital	5,133.42	6,163.56	6,315.00	11,335.00	12,733.00
% of Revenue	27%	31%	31%	42%	33%

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Definition

Free Cash Flow (FCF) =

Gross Cash Flow – Gross Investment

=NOPAT – Net Investment

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Free Cash Flow of Titan

	2019	2020	2021	2022	2023
Other Non-Interest Bearing Liabilities	118.85	152.46	143.00	179.00	214.00
Increase in NIBL		33.61	-9.46	36.00	35.00
Capex		1,111.16	226.44	343.00	639.00
Increase in Operating Wkg Cap		1,030.14	151.44	5,020.00	1,398.00
Gross Investment		2,107.69	387.34	5,327.00	2,002.00
Free Cash Flow		**-552.29**	**592.66**	**-3,160.05**	**1,298.43**

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Definitions

Operating Invested Capital (OIC) = Operating FA + Operating Wkg. Capital + Other Operating Assets

Essentially operating invested capital is used to generate the NOPAT for a company.

$$ROIC = \frac{NOPAT}{Operating \quad Invested \quad Capital} \quad ...5(14)$$

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Definitions

EVA =

Operating Invested Capital × (ROIC – WACC)

$$OIC \times (ROIC - WACC) = OIC \times \left(\frac{NOPAT}{OIC} - WACC\right) = NOPAT - OIC \times WACC.$$

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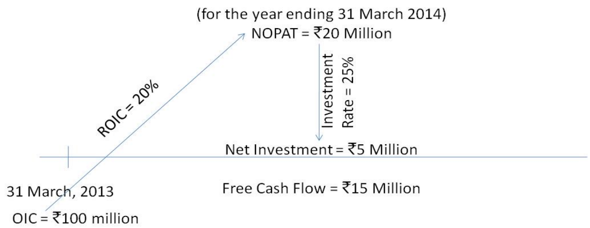
Growth Rate…

- OIC as on 31 March 2013 = ₹100m
- ROIC = 20%
- Investment rate = 25%

Investment Rate = Net Investment / NOPAT



g = ROIC × Investment Rate

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Valuation using EVA Method

- OIC as on 31 March 2022 = ₹100m
- ROIC = 20%
- Investment rate = 25%
- What is the enterprise value?
- What will be the value if the growth rate is 0?
- What is the value, if the growth rate is 10%?

Value=OIC+PV(EVA)

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Where to Invest?

- Operating Invested Capital = Rs.100 million
- Cost of Capital = 15%
- Projected Growth Rate = 5%
- The ROIC of
  - Company A is 10%
  - Company B is 15%
  - Company C is 20%
- Where will you invest?

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Value Using EVA Method

Year	0	1	2	3	4	5
OIC	100	113	125	135		
ROIC	20%	20%	20%	20%	20%	20%
NOPAT		20	22.6	25		
Net Investment		13	12	10		
Free Cash Flow		7	10.6	15		
WACC		13%	13%	13%		
EVA		7	7.91	8.75		
terminal Growth I	5%					

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Valuation Using EVA Method

Year	0	1	2	3	4	5
OIC	100	113	125	135		
ROIC	20%	20%	20%	20%	20%	20%
NOPAT		20	22.6	25	27	
Net Investment		13	12	10	6.75	
Free Cash Flow		7	10.6	15	20.25	
WACC		13%	13%	13%	13%	
EVA		7	7.91	8.75	9.45	
terminal Growth I	5%					
PV(EVA)		6.19	6.19	6.06	81.87	
Value-EVA	200.32					
PV(FCF)		6.19	8.30	10.40	175.43	
Value-FCF	200.32					

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